



LINUX FUNDAMENTALS

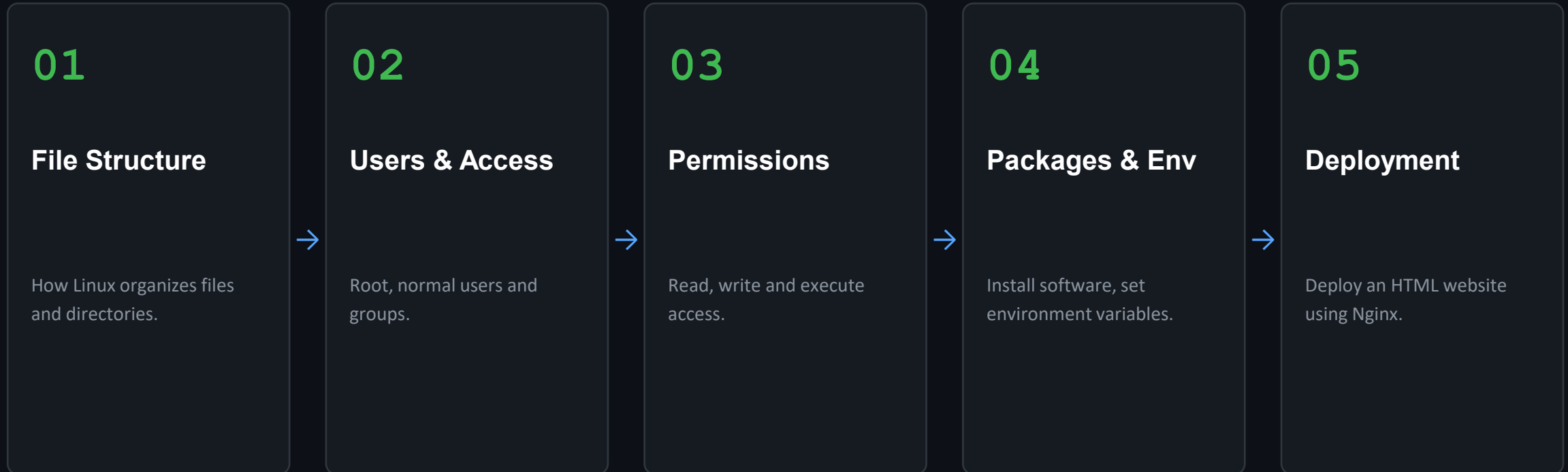
File Structure • Users • Permissions • Packages • Environment Variables • Website Deployment



```
$ whoami  
root  
$ ls  
$ sudo apt install nginx
```

Today's Learning Objectives

A practical roadmap from Linux basics to a live website



Linux File Structure

Everything starts from the root directory /

Linux follows a **hierarchical filesystem**. All files and directories begin from a single root, written as

**TIP**

Think of / as the starting point of the entire Linux filesystem.

```
/
├── home
│   ├── rahul
│   └── amit
├── root
├── etc
├── var
├── usr
├── tmp
├── boot
├── bin
└── dev
```

Important Linux Directories

The directories every beginner should recognize first

DIRECTORY	PURPOSE
/	Root of the entire filesystem
/home	Personal directories of normal users
/root	Home directory of the root user
/etc	System configuration files
/var	Frequently changing data, logs & app data
/tmp	Temporary files
/usr	Applications, libraries and utilities
/bin	Essential commands
/boot	Boot-related files
/dev	Device files

Focus on these directories first — the rest can be learned gradually as you go.

Linux Users — Root User

The superuser / administrator account

The **root user** is the administrator of a Linux system and has very high privileges over the entire machine.



ROOT USER

System Administration

/ → Root directory of Linux

/root → Home directory of root user



```
$ whoami  
root
```

⚠ WARNING

Root has powerful access that can affect the whole system. Use administrative privileges carefully.

Other / Normal Users

Everyday accounts used for daily work

A normal user is an account created for everyday work on Linux — writing files, running programs, browsing the system.

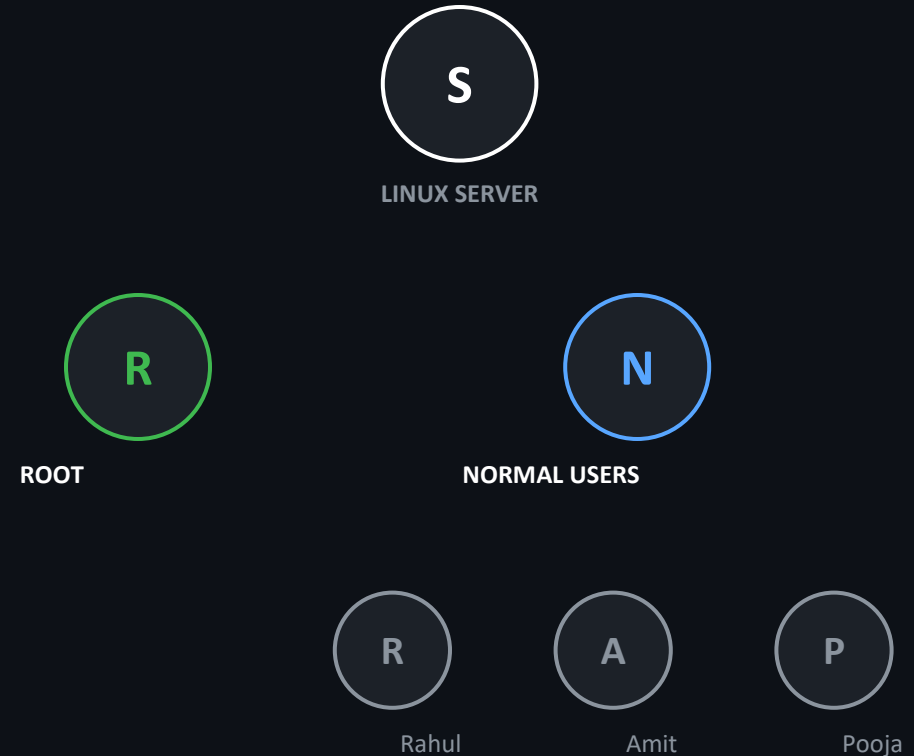


```
/home/rahul  
/home/amit  
/home/pooja
```



```
$ whoami  
$ sudo groupadd gropuname  
$ sudo usermod -aG gropuname username  
$ groups username
```

Normal users can work with files and applications where they have permission.



Root User vs Normal User

A side-by-side comparison of privileges

FEATURE	ROOT USER	NORMAL USER
Purpose	System administration	Daily work
Privileges	Very high	Limited
Home directory	/root	/home/username
Create users	Yes	Normally no
Install system packages	Yes	Usually requires sudo
Modify system config	Yes	Usually restricted
Manage services	Yes	Usually requires privileges
Access protected files	Generally yes	Depends on permissions

IMPORTANT sudo gives an authorized user elevated privileges for a single command.

Linux File Permissions

Who can read, modify or execute a file?

Linux permissions control access to files and directories using three permission types and three categories of user.



Reading Linux Permissions

Breaking down the output of `ls -l`

```
$ ls -l
-rwxr-xr-x webapp
```

`ls -l` is one of the most useful commands for understanding file permissions.

—
|
File type

rwX
|
Owner - u

r-x
|
Group- g

r--
|
Others - o

Owner → **rwX** Group → **r-x** Others → **r--**

Environment Variables

Configuration that lives outside your application code

An environment variable stores information as a **key-value** pair:

```
NAME=VALUE

PORT=8000
APP_NAME=MyApp
DATABASE_URL=...
```

```
$ env
$ printenv
$ echo $HOME
$ echo $PATH
```

Linux Environment

```
|— PORT=8000
|— APP_NAME=MyApp
└— DATABASE_URL=...
```



APPLICATION

PATH

PATH tells the shell which directories to search when looking for executable commands.

Environment Variables in Practice

Setting a variable in Linux and reading it from Python



```
$ export APP_NAME="MyApplication"
$ echo $APP_NAME
MyApplication
```



```
import os

app_name = os.getenv("APP_NAME")

print(app_name)
```

SECURITY NOTE

Do not commit passwords, API keys or secret keys to public repositories.

Linux



APP_NAME=MyApplication



Python



os.getenv("APP_NAME")



MyApplication

Deploy a Website on Linux

HTML → Nginx → Browser



```
$ sudo apt update
$ sudo apt install nginx
$ cd /var/www/html
$ sudo vim /var/www/html/index.html
$ sudo systemctl start nginx
```



```
<!DOCTYPE html>
<html>
<head><title>My Linux Website</title></head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This website is running on Linux.</p>
</body></html>
```

Browser



Nginx



/var/www/html



index.html

`/var/www/html` is a common default web root for the default Nginx site on Ubuntu/Debian.

Linux Practical — From Server to Website

Recap workflow and hands-on task

Linux Server



Create User



Understand Permissions



Install Nginx



/var/www/html



Create index.html



Start / Check Nginx



Find Server IP



LIVE WEBSITE



```
$ whoami
$ ls -l
$ chmod 755 script.sh
$ sudo apt update
$ sudo apt install nginx
$ sudo systemctl status nginx
$ cd /var/www/html
$ sudo vim index.html
$ hostname -I
```

STUDENT PRACTICE TASK

1. Check user
2. Explore /var /home /etc
3. Check permissions
4. Install Nginx
5. Create index.html
6. Copy to /var/www/html
7. Check Nginx status
8. Find server IP & open in browser

“From understanding Linux → to deploying your first website.”